



Investigation of ion-exchange ‘stuffed’ glass structures by molecular dynamics simulation

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ABSTRACT

Molecular dynamics simulations were used to examine ion-exchange ‘stuffed’ glass structures by substitution of Na^+ ions with K^+ ions in $x\text{Na}_2\text{O} \cdot (100 - x)\text{SiO}_2$, $x = 9, 16$, and $23 \text{ mol\%}$. After substitution, relaxation was performed under varied boundary conditions, including NVT and NPT ensembles. Structural features of stuffed glasses were compared with those of the host glass and of the as-formed, compositionally-equivalent potassium silicate glass. Stuffing K^+ ions had oxygen coordination number near that of the as-formed potassium silicate. Ion-exchange stuffed glasses were topologically confined, showing little Q^n distribution or ring size distribution change compared to the host glass. Intertetrahedral re-orientation occurred to bear some of the network strain. Resulting molar volume change of ion-exchange stuffed systems was 58–67% of the difference between the as-formed sodium silicate host and the as-formed end-member potassium silicate glass. This accounted for a portion of the reduced experimental surface compression observed during ion exchange relative to that expected from the linear elastic suppression of the difference of as-melted molar volumes.

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1. Introduction

Ion-exchange strengthening of glass introduces compression into glass surface layers to mitigate the effects of tensile stress-concentrating flaws. This technology is traditionally found in aircraft cockpit windows, photocopier windows, and auto-injector cartridges for time-sensitive medication delivery. More recent applications include high-RPM hard disk platter substrates and protective display windows in mobile electronic devices. Demand for this technology is likely to increase as glass is further incorporated into products where high strength, high scratch resistance, and/or handling tolerance are required. Ion-exchange strengthening relies upon the interdiffusion of alkali ions between an alkali-containing host glass and an alkali salt bath. When the salt bath contains alkali ions of larger ionic radius than those in the glass and when the process is carried out below the glass strain point, compression develops in the ion-exchanged surface layers of the glass. Surface compression magnitudes of 200 to 1000 MPa can be generated as the larger alkali ions ‘stuff’ into host sites. The kinetics of the interdiffusion process are well established [1,2]. In soda lime silicates, invading potassium ions can extend to depths of typically 20–50 μm from the surface. The resulting compression magnitude is roughly proportional to the concentration of stuffing alkali. Thus, depth of compression nearly corresponds to the depth of stuffing alkali diffusion and interior tension develops to

balance this surface compression. Although chemical strengthening technology has been known for nearly 50 years, atomistic mechanisms ultimately responsible for compression generation are not well-understood [2].

Stress generation by ion-exchange stuffing can be examined analogously to thermal stress generation. A linear network dilatation coefficient (LNDC) [3], comparable to a linear thermal expansion coefficient, represents the linear strain per quantity stuffing alkali species. Linear network dilatation coefficient, B , is defined as [4]:

$$B = \frac{1}{3} \left(\frac{1}{V} \frac{\partial V}{\partial C} \right) \quad (1)$$

where V is molar volume and C is concentration stuffing alkali species. A typical approximation [5] to understanding ion-exchange strengthened glasses is to examine the molar volumes of the host glass, for example soda lime silicate, and an as-melted glass of equivalent composition to the surface layer of the strengthened glass, for example potash lime silicate. If a potash lime silicate surface layer was elastically constrained to a soda lime silicate host, the expected resulting compression within the surface layer is $\sim 3.5 \text{ GPa}$ ($\sim 2.5 \times 10^{-3}/(\text{mol\% K}_2\text{O})$ in terms of LNDC), whereas laboratory-determined compression rarely exceeds 500–600 MPa ($\sim 4.0 \times 10^{-4}/(\text{mol\% K}_2\text{O})$ in terms of LNDC) for chemically strengthened soda lime silicate glass. This discrepancy remains after accounting for viscous relaxation and has been noted by a number of researchers [4,5]. Properties anticipated to influence the stress state of stuffed layers, such as Young’s modulus, Poisson’s ratio, and thermal

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expansion coefficient are altered by ion exchange, but these changes are thought to be far from that needed to account for the compression magnitude discrepancy [6,7]. Clearly the structure of ion-exchanged layers is significantly different than that of the compositionally-equivalent as-melted structure [5,6]. This is evident in measurements of stuffed layer density for lithium replaced by stuffing sodium in aluminosilicate glass. Ion-exchange stuffed layers are denser than compositionally-equivalent as-melted glass [8]. The question remains as to how stuffing alkali ions are accommodated by the silicate network.

The present study utilized molecular dynamics simulation to provide insight into structural features of ion-exchange 'stuffed' glasses. Three sodium silicate glass compositions were simulated, substituting K^+ in place of Na^+ in various quantities then the resulting structural changes were observed under NVT and NPT ensembles. The stuffed glasses developed lower molar volumes than those of compositionally-equivalent as-formed potassium silicate glasses due to the inability of the silicate network to undergo topological rearrangement.

2. Simulation method

Three glass compositions were studied, $xNa_2O \cdot (100-x)SiO_2$ where $x=9, 16$, and 23 mol%, along with their potassium silicate compositional equivalents. For each sodium silicate composition, a 100% K^+ ion substitution for Na^+ was performed and the systems were relaxed under various boundary conditions. Additionally, for each sodium silicate composition, fractional K^+ ion substitution for Na^+ at 25%, 50%, and 75% of the host Na_2O concentration was performed under NPT ($P=0$, $T=300$ K) conditions. Simulation details were as follows. Glasses were initially formed by random placement of 6000 atoms in a cubic simulation box with length selected to target molar volume of compositionally-equivalent laboratory glasses, where laboratory values were estimated from a mean of three to ten SciGlass database [9] entries found to be obtained under appropriate forming and measurement conditions. Simulations were conducted using the DL_POLY software package [10] with cubic periodic boundary conditions using potentials developed by Pedone et al. [11]. Simulations under NVT ($T=6000$ K) were performed for 40 ps, followed by cooling to 300 K at 5 K/ps in intervals of -0.01 K/time step. Following this, relaxation for 100 ps under NPT ($P=0$, $T=300$ K) conditions generated what is referred to in this work as 'as-formed' or 'end-member' glasses. For all simulations, the time step increment was 2 fs and electrostatic interactions were calculated by the Ewald sum technique [12].

Ion substitution was accomplished by relabeling appropriate quantities of Na^+ ions to K^+ ions, then allowing the system to relax a further 200 ps under either NPT ($P=0$) or NVT, with $T=300$ K. The Na^+ ions replaced by K^+ ions were randomly selected and no attempt was made to model chemical potential gradients or interdiffusion processes. The $23Na_2O \cdot 77SiO_2$ system, after K^+ ion substitution, was additionally run under NPT ($P=350$ MPa, $T=300$ K) conditions, where the system pressure was selected to mimic an estimated maximum achievable surface compressive stress, determined in our unpublished work of K^+ for Na^+ ion exchange in $23Na_2O \cdot 77SiO_2$ laboratory glass. The simulation temperature was selected to be well below that of a typical laboratory ion-exchange strengthening process temperature (600–800 K), in an attempt to inhibit temperature-activated relaxations, thereby observing structural features involved in the immediate stages of accommodation of the stuffing alkali ions by the silicate network. The simulation naming convention for as-formed simulations is $xMSilicate$, where x = mol% M_2O and $M=Na$ or K , and for ion-exchanged simulations is $x_Swap(\#)_nvt$ or $x_Swap(\#)_npt_y$, where x has been defined, $(\#)$ represents the fraction of Na^+ ions replaced by K^+ ions (25, 50, 75, or 100%) and y is the specified system pressure in units of MPa.

To briefly examine the influence of ion-exchange temperature, an as-formed $23Na_2O \cdot 77SiO_2$ simulation was brought to 623 K under NVT ensemble at a rate of 0.01 K/time step (5 K/ps), then relaxed

for 200 ps under NPT, $P=0$, $T=623$ K. Ion exchange was performed as described earlier, where after 100% exchange of Na^+ ions to K^+ , the system was allowed to proceed 200 ps under NPT ($P=0$, $T=623$ K) before being cooled to 300 K under NVT ensemble at a rate of -0.01 K/time step (5 K/ps). Once at 300 K, a relaxation under NPT ($P=0$, $T=300$ K) for 100 ps was performed. After this final relaxation, the resulting system was labeled 23_Swap100_npt_0_623K.

To investigate whether 'stuffing' ion exchange causes permanent structural deformation, after K^+ ion substitution and silicate network accommodation, a set of the NPT ($P=0$, $T=300$ K) systems ($x_Swap100_npt_0$) had Na^+ ions substituted in place of K^+ ions and the structures were again relaxed under NPT ($P=0$, $T=300$ K) to generate the simulations labeled $x_SwapBack100_npt_0$, where x has been defined. A summary of the simulations is provided in Table 1.

Properties of the ion-exchanged stuffed and the back-exchanged simulated glasses were compared to as-formed sodium silicate and potassium silicate simulated glasses. All properties were determined from the final 20 ps of each simulation, sampled in 0.1 ps intervals. Mean coordination numbers, Q^n distributions, bond angle distributions, and ring size distributions were calculated using a cut-off length corresponding to the first minimum in the pair distribution functions. For silicon-oxygen pairs, this corresponded to a cut-off length of 2.20 Å for all simulations. For alkali-oxygen pairs, the cut-off length was dependent upon the alkali species and simulation type (as-formed or 'stuffing'), therefore the cut-off length is included with the mean coordination number in Table 2.

3. Results

3.1. Ion exchange at 300 K

Alkali-oxygen mean coordination numbers (CN), where the alkali ion is the central atom surrounded by oxygen (i.e., (M)O), are given in Table 2. The reported error is one standard deviation of the mean. The as-formed glasses had (Na)O and (K)O mean CN of ~ 6 and ~ 8 , respectively, and agree reasonably well with those reported in literature

Table 1
Summary of the molecular dynamics simulations examined.

Simulation	Ensemble	System pressure (MPa)	Stuffing K_2O (mol%)
9NaSilicate	NPT	0	0, as-formed
9KSilicate	NPT	0	0, as-formed
9_Swap100_nvt	NVT	Resulting 900	9.0
9_Swap025_npt_0	NPT	0	2.25
9_Swap050_npt_0	NPT	0	4.5
9_Swap075_npt_0	NPT	0	6.75
9_Swap100_npt_0	NPT	0	9.0
9_SwapBack100_npt_0	NPT	0	0, stuffing K^+ removed
16NaSilicate	NPT	0	0, as-formed
16KSilicate	NPT	0	0, as-formed
16_Swap100_nvt	NVT	Resulting 2000	16.0
16_Swap025_npt_0	NPT	0	4.0
16_Swap050_npt_0	NPT	0	8.0
16_Swap075_npt_0	NPT	0	12.0
16_Swap100_npt_0	NPT	0	16.0
16_SwapBack100_npt_0	NPT	0	0, stuffing K^+ removed
23NaSilicate	NPT	0	0, as-formed
23KSilicate	NPT	0	0, as-formed
23_Swap100_nvt	NVT	Resulting 3,300	23.0
23_Swap025_npt_0	NPT	0	5.75
23_Swap050_npt_0	NPT	0	11.5
23_Swap075_npt_0	NPT	0	17.25
23_Swap100_npt_0	NPT	0	23.0
23_Swap100_npt_350	NPT	Specified 350	23.0
23_Swap100_npt_0_623K ^a	NPT	0	23.0
23_SwapBack100_npt_0	NPT	0	0, stuffing K^+ removed

^a Ion-exchange stuffing carried out at 623 K. Structural analysis performed at 300 K.

Table 2

Mean alkali-oxygen coordination numbers, mean intertetrahedral angles, molar volumes, and linear network dilatation coefficients.

Simulation	(M)O, mean coordination Number [Cut-off length in Å]	Mean Si–O–Si Angle (degrees)	Molar volume (cm ³ /mol)	LNDC (per mol% K ₂ O) From sodium silicate host
9NaSilicate	Na, 5.7 ± 1.4 [3.28]	152.0	25.79	–
9KSilicate	K, 7.6 ± 2.1 [3.73]	152.2	26.82	1.49E–03
9_Swap100_nvt	K, 8.2 ± 1.7 [3.63]	151.6	25.78	–
9_Swap025_npt_0	K, 8.7 ± 1.9 [3.75]; Na, 5.6 ± 1.4 [3.26]	152.1	25.93	7.99E–04
9_Swap050_npt_0	K, 7.9 ± 1.7 [3.62]; Na, 5.7 ± 1.4 [3.28]	152.3	26.08	8.35E–04
9_Swap075_npt_0	K, 8.4 ± 1.8 [3.72]; Na, 5.3 ± 1.3 [3.19]	152.5	26.24	8.73E–04
9_Swap100_npt_0	K, 8.3 ± 1.8 [3.69]	152.6	26.38	8.57E–04
9_SwapBack100_npt_0	Na, 5.7 ± 1.4 [3.28]	151.9	25.77	–
16NaSilicate	Na, 5.7 ± 1.3 [3.23]	151.8	25.11	–
16KSilicate	K, 8.0 ± 2.0 [3.81]	152.1	27.27	1.79E–03
16_Swap100_nvt	K, 8.4 ± 1.7 [3.65]	151.1	25.13	–
16_Swap025_npt_0	K, 9.1 ± 1.7 [3.79]; Na, 5.5 ± 1.3 [3.22]	152.1	25.42	1.04E–03
16_Swap050_npt_0	K, 8.7 ± 1.7 [3.75]; Na, 5.4 ± 1.3 [3.21]	152.6	25.84	1.21E–03
16_Swap075_npt_0	K, 8.4 ± 1.8 [3.74]; Na, 5.5 ± 1.3 [3.24]	152.9	26.20	1.21E–03
16_Swap100_npt_0	K, 8.4 ± 1.8 [3.75]	153.1	26.46	1.12E–03
16_SwapBack100_npt_0	Na, 5.7 ± 1.3 [3.23]	151.7	25.08	–
23NaSilicate	Na, 5.8 ± 1.2 [3.25]	151.3	24.58	–
23KSilicate	K, 7.4 ± 1.7 [3.75]	152.2	27.77	1.88E–03
23_Swap100_nvt	K, 7.9 ± 1.5 [3.55]	150.2	24.55	–
23_Swap025_npt_0	K, 8.0 ± 1.5 [3.64]; Na, 5.5 ± 1.2 [3.22]	151.7	25.14	1.33E–03
23_Swap050_npt_0	K, 8.4 ± 1.6 [3.75]; Na, 5.7 ± 1.3 [3.30]	152.2	25.68	1.29E–03
23_Swap075_npt_0	K, 8.4 ± 1.7 [3.77]; Na, 5.5 ± 1.1 [3.28]	152.7	26.18	1.26E–03
23_Swap100_npt_0	K, 7.5 ± 1.6 [3.65]	153.1	26.72	1.26E–03
23_Swap100_npt_350	K, 7.7 ± 1.6 [3.65]	152.8	26.50	1.13E–03
23_Swap100_npt_0_623K	K, 7.8 ± 1.6 [3.71]	152.9	26.82	1.32E–03
23_SwapBack100_npt_0	Na, 5.8 ± 1.2 [3.25]	151.2	24.54	–

determined via EXAFS and simulation [13–17]. NVT stuffed glasses had mean (K)O CN greater than that of the end-member potassium silicate. NPT stuffed glasses displayed mean (K)O CN similar to their as-formed potassium silicate end-members for all three simulated glass series, tending toward slight over-coordination. This suggests that the coordination environment around the stuffing K⁺ ion is similar to that in the as-formed potassium silicate, at least with regard to number.

Network connectivity, inspected by Q^n speciation where n represents the number of bridging oxygen associated with a SiO₄ tetrahedron, showed small differences between the as-formed sodium and potassium silicates. For as-formed sodium silicate simulations, Q^n values as percentages of the total silicon count were Q^2 : 0.6%, 2.7%, 8.9%, Q^3 : 18.3%, 32.3%, 40.5%, and Q^4 : 80.8%, 64.8%, 50.0% for $x=9$, $x=16$, and $x=23$ compositions, respectively. Q^n distributions for as-formed potassium silicates were within 0.6% of each of these values. These Q^n values were in good agreement with the literature [18,19]. Little to no net difference of Q^n species between the stuffed simulations (regardless of NVT or NPT ensembles) and their host glasses were observed. This is consistent with the findings of Tandia et al. [20]. Ring size distributions (RSD), where ring size is defined as the number of silicon atoms contained in a complete ring, differed between the as-formed sodium and potassium silicate glasses (Fig. 1) and the stuffed simulations exhibited distributions very similar to their hosts. Unchanged RSD after ion-exchange stuffing indicates an inability of the silicate network to achieve the topological configuration of the as-formed potassium silicate structure. That is, although stuffing potassium ions occupy former sodium ion sites, the sodium silicate network topology of the host remains intact.

The mean tetrahedral bond angle (O–Si–O) was consistent, at 109°, across all simulations and that value is well accepted in literature [11,14]. This suggests the shape of the SiO₄ tetrahedra is maintained upon stuffing alkali accommodation. Mean intertetrahedral bond angles (Si–O–Si) varied slightly between the as-formed sodium and potassium silicate glasses (Table 2), the largest difference being 0.9° for the $x=23$ mol% composition. The Si–O–Si bond angle of 150–152° is in agreement with sodium silicate simulations of Pedone et al. [11], but is larger than that deduced in a recent combined NMR

and molecular dynamics simulation study by Angeli et al. [21], where a 22.5Na₂O·77.5SiO₂ glass was found to have a mean Si–O–Si angle of 141.7°. Pedone et al. [11] attributed the large Si–O–Si angle to limited simulation size and lack of polarizability factors in computations. The NVT stuffed systems showed a slight contraction of the Si–O–Si bond angle to 0.4–1.0° less than that of the host glass. For all NPT fully stuffed systems, the Si–O–Si bond angle exceeded that of the as-formed potassium silicate end-member, the magnitude varying from 0.4° for $x=9$ mol% to ~1° for $x=16$ and 23 mol%. Si–O–Si bond angles exceeding that of their potassium end-members is representative of the participation of intertetrahedral re-orientation upon silicate network accommodation of stuffing potassium ions.

NVT stuffed simulations generated 0.9, 2.0, 3.3 GPa system pressure for $x=9$, 16, and 23 mol%, respectively, after complete exchange of K⁺ for Na⁺. These values are significantly higher than that found in laboratory ion-exchanged glasses. For example, our unpublished work of K⁺ for Na⁺ ion exchange in a 23Na₂O·77SiO₂ laboratory glass generated less than 350 MPa surface compression. This suggests the structures of the NPT stuffing simulations may be more appropriate representations of laboratory glasses, especially in light of the closeness of the properties in Table 2 between NPT ($P=0$) and NPT ($P=350$ MPa) systems.

The molar volumes (Table 2) of the simulated end-member glasses were within ~3% or better of their target values, and the simulated glasses had lower molar volume than that reported in the literature. Each simulation had molar volume error of 0.02 cm³/mol based upon the standard deviation of the simulation volume across the averaged time steps. The closed symbols in Fig. 3 represent the molar volumes of as-formed glasses and the open symbols represent molar volumes of NPT stuffed glasses. Molar volume increased linearly with stuffing alkali concentration. At 100% exchange of K⁺ replacing Na⁺, the resulting molar volume change accounted for only 58%, 63%, and 67% of the difference between the as-formed end-members, for $x=9$, 16, and 23 mol%, respectively. This, in turn, produced linear network dilatation coefficients (LNDC) less than that expected from molar volume differences of the as-formed end-member glasses. For example, the LNDC computed from molar volume differences of the $x=23$ mol% as-

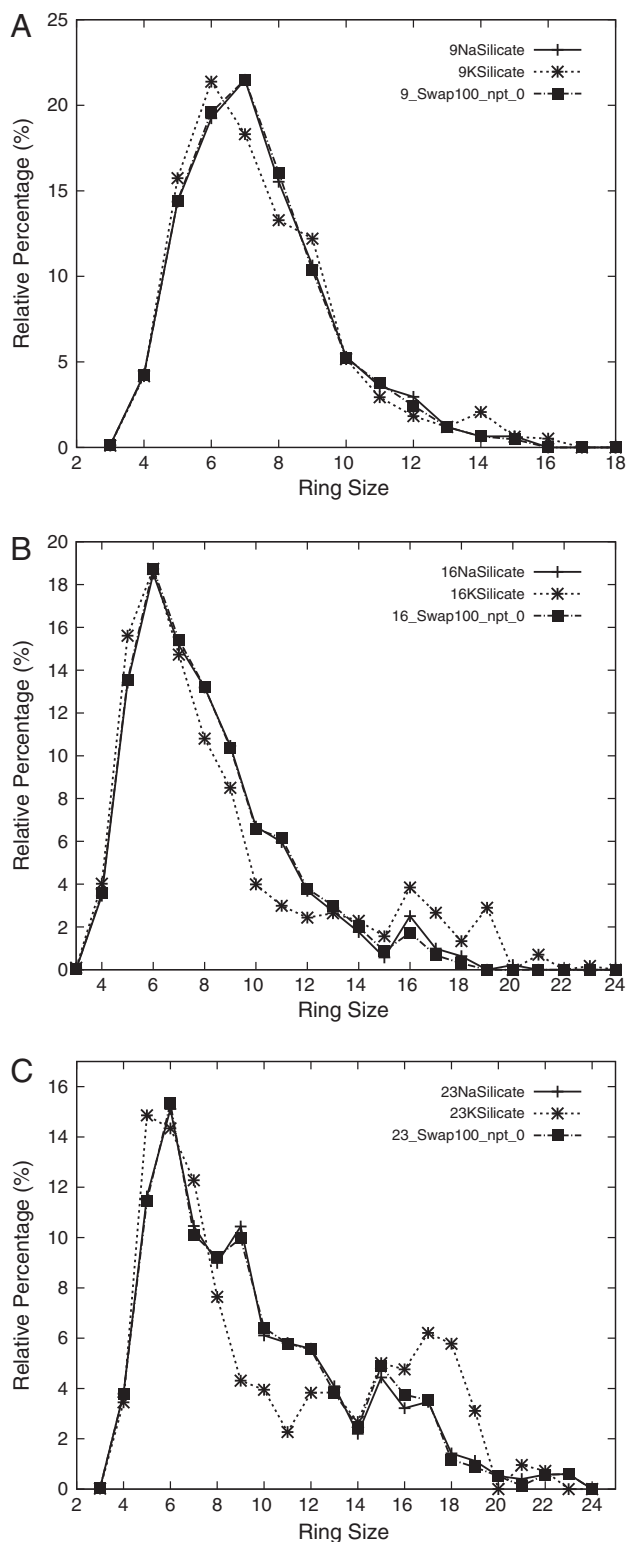


Fig. 1. Ring size distributions for (A) $x=9$ mol%, (B) $x=16$ mol%, and (C) $x=23$ mol% end-member and ion-exchange simulated glasses. Ion-exchange stuffing is carried out at 300 K and the structure is analyzed at 300 K. Note, traces for $x\text{NaSilicate}$ and $x_{\text{Swap100_npt_0}}$ largely overlap for all ring sizes.

formed end-members is $1.88 \times 10^{-3}/(\text{mol}\% \text{K}_2\text{O})$, whereas the fully K^+ substituted NPT ($P=0$) system has a value of $1.26 \times 10^{-3}/(\text{mol}\% \text{K}_2\text{O})$. LNDC values of NPT stuffed simulations are given in Fig. 4 versus molar percentage stuffing K_2O . LNDC errors were calculated accounting for molar volume error and the molar percentage stuffing K_2O . The glass

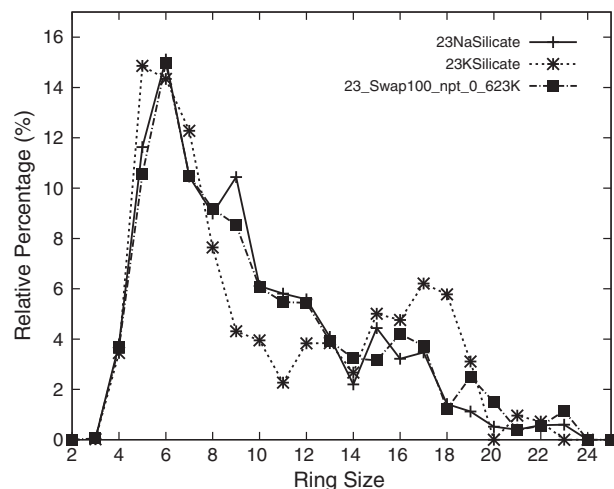


Fig. 2. Ring size distributions for $x=23$ mol% end-member and ion-exchange simulated glasses. Ion-exchange stuffing is carried out at 623 K and the structure is analyzed at 300 K. Note, traces for 23NaSilicate and $23_{\text{Swap100_npt_0_623K}}$ largely overlap for all ring sizes.

series $x=9$ and 23 mol% had somewhat constant LNDC values with increasing percentage of exchange, whereas for $x=16$ mol% a shallow maximum around 50 to 75% exchange was seen.

Despite the present stuffed simulations having generated a lower molar volume than their end-member potassium silicates, if the $x=23$ mol%, fully K^+ exchanged NPT ($P=0$) glass structure was elastically confined to its sodium silicate host structure, the resultant stress would be ~ 2.2 GPa (assuming Young's modulus of 60 GPa and Poisson's ratio of 0.2), which is more than half an order of magnitude greater than that observed in laboratory glass. On the other hand, the difference between LNDC of $x=23$ mol%, fully K^+ stuffed NPT ($P=0$) and NPT ($P=350$ MPa) systems is $1.31 \times 10^{-4}/(\text{mol}\% \text{K}_2\text{O})$, which represents the linear elastic incompatibility required to place the $P=0$ system under 350 MPa of system pressure. If one assumes a laboratory glass in which 23 mol% K_2O replaces Na_2O and this exchange generates 350 MPa surface compression with Young's modulus of 60 GPa and Poisson's ratio of 0.2, the resultant LNDC is $\sim 2.0 \times 10^{-4}/(\text{mol}\% \text{K}_2\text{O})$. The similarity of the simulated elastic LNDC and the simple laboratory-estimated elastic LNDC suggests the elastic behavior of the simulation is not unlike that of the laboratory glass.

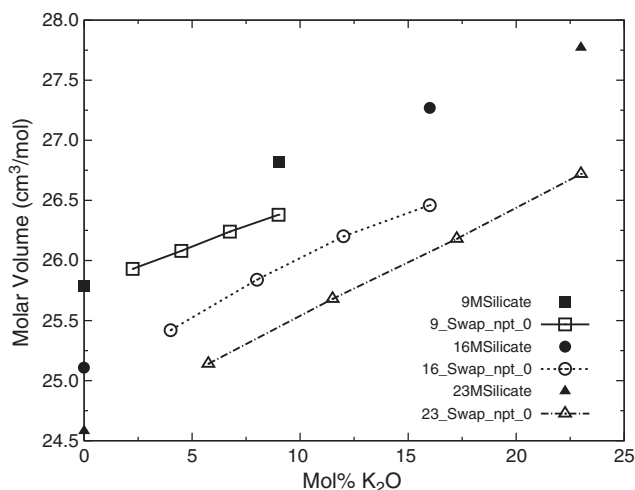


Fig. 3. Molar volumes of as-formed and ion-exchange simulated glasses. Data points at 0 mol% K_2O are for as-formed sodium silicate systems ($x\text{NaSilicate}$), where $x=23$ and $x=9$ have the lowest and highest molar volumes, respectively. Molar volume error bars are within the symbol height.

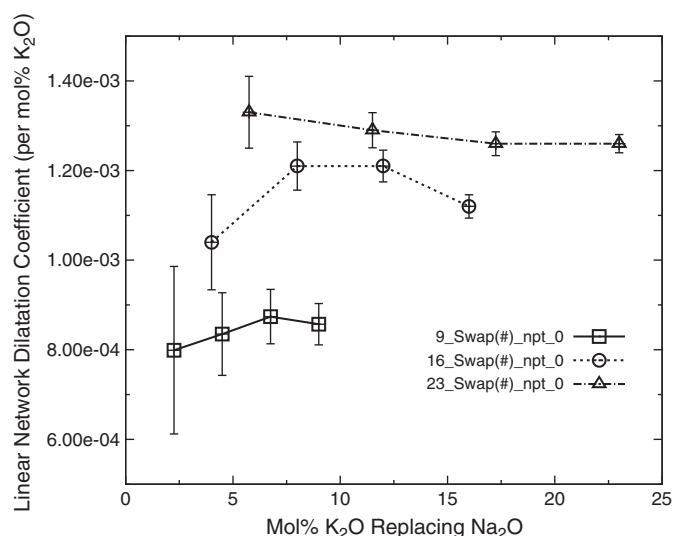


Fig. 4. Linear network dilatation coefficients for ion-exchange simulated glasses.

3.2. Ion exchange at 623 K

Performing ion exchange at temperature of 623 K and then examining structural properties at 300 K (23_Swap100_npt_0_623K), showed few differences from the ion-exchange simulations carried out at 300 K (23_Swap100_npt_0). Mean K(O) CN and mean Si–O–Si bond angle were similar between the two K⁺ stuffing simulations performed at different temperatures (Table 2). Little net difference of Q^n species was observed between the 623 K ion-exchange simulation and the as-formed host sodium silicate glass. RSD of the 623 K ion-exchange simulation (Fig. 2) showed a bit greater variation from the as-formed host glass than the 300 K ion-exchange simulation, but remained close to that of the host glass, as observed earlier (Fig. 1C). When examined at 300 K, the molar volume and LDNC of the 623 K ion-exchange and 300 K ion-exchange simulations were also similar, taking on slightly higher values for 623 K ion exchange.

3.3. Back exchange of the 300 K ion-exchange simulations

Back-exchange (x_SwapBack100_npt_0) simulations, where the stuffed K⁺ ions were removed and replaced by Na⁺ ions, showed restoration of the mean (Na)O coordination number, the mean Si–O–Si bond angle, and molar volume to that of the as-formed host for all three compositions (Table 2). Therefore, the observed structural changes by network accommodation of stuffing alkali are nearly completely elastic. Careful inspection of Si–O–Si bond angle and molar volume found slight differences of 0.1° lower Si–O–Si bond angle and 0.1–0.4 cm³/mol lower molar volume for the back-exchanged simulated glasses. These differences were within the error of the measurements, but the consistency of the trend between three different glass compositions allows for the possibility of subtle permanent deformation of the silicate network caused by accommodation of stuffing alkali ions.

4. Discussion

The present simulation set indicates that the silicate network accommodates stuffing K⁺ ions by the following process. A K⁺ ion replaces a Na⁺ ion at a host alkali site and the ionic size mismatch drives a limited expansion of the site. Site expansion and the K⁺ ion's desired chemical coordination draw additional oxygen atoms toward the site, increasing the surrounding oxygen count from ~6 to ~8. Due to the displacement of oxygen atoms, both from alkali site expansion and those pulled toward the exchanged site, the silicate

network is left strained. This strained state is evident, in part, by re-orientation of tetrahedral SiO₄ units with respect to one another, indicated by an increase in the Si–O–Si bond angle. With sufficient quantity of stuffing K₂O, the Si–O–Si bond angle exceeds that of the as-formed potassium silicate end-member, further highlighting the role of intertetrahedral re-orientation during network accommodation of stuffing alkali. Insufficient thermal energy is present to allow for net bond breaking and reformation to relax the strained state of the network, within the time scales explored by these simulations, leaving the Q^n and ring size distributions unchanged, and limiting the expansion of the network due to topological confinement. The resulting ion-exchanged 'stuffed' glass structure has local (K)O coordination number similar to the as-formed potassium silicate structure, but has the topological configuration of the sodium silicate host structure. As a consequence, the molar volume of the potassium-stuffed glass is less than that of the as-formed (as-melted), compositionally-equivalent potassium silicate.

This suggests the following implications for laboratory glasses. There is a maximum achievable molar volume produced by the ion-exchange process, which is less than that of the compositionally-equivalent, as-melted molar volume. This provides a limit to the achievable beneficial compression that can be developed. Thermal activation could enable relaxation of the ion-exchange stuffed network structures, leading to greater molar volume as the structure relaxes toward that of the as-melted glass, but this relaxation would be concurrent to stress relaxation. Therefore relaxation of the stuffing alkali accommodation features is likely to relax the beneficial compression in the ion-exchanged layers. These points are in agreement with the recent findings of Tandia et al. [20], although here the topological configuration, rather than alkali under-coordination by oxygen, is the structural feature limiting the achievable molar volume.

The molar volume of the simulated ion-exchange stuffed glasses in the present work is still greater than that suggested by the stress state measured in laboratory glasses. As a first approximation to model ion-exchange strengthened glasses, the present simulations utilized cubic periodic boundary conditions, which resulted in equal expansion in three dimensions after ion-exchange stuffing. Laboratory ion-exchange strengthened glasses most often have anisotropic boundary conditions that are better modeled by constrained expansion in two dimensions (in-plane) and free expansion in the third dimension. The incompatible expansion in the constrained, in-plane dimensions differs from the expansion in the third, free dimension [5,22]. Measurements by Burggraaf [5] of differential swelling between K⁺ ion-exchange strengthened and non-ion-exchanged portions of the same freely expanding surface of a 22.15Na₂O·7.55Al₂O₃·70.3SiO₂ laboratory glass showed the free expansion to be on average on the order of 10^{−3}/(mol% K₂O) (in terms of LNDC) versus an in-plane incompatible expansion, back-calculated from the surface compression state, on the order of order of 10^{−4}/(mol% K₂O). The present simulated glasses have similar order of magnitude free LNDC and incompatible LNDC (recall, the incompatible LNDC is the LNDC difference between simulations 23_Swap100_npt_0 and 23_Swap100_npt_350), but are performed under isotropic boundary conditions. This difference of boundary conditions is possibly responsible for the stress of ~2.2 GPa calculated between the x = 23 mol% ion-exchange stuffed and as-formed sodium silicate host simulations, if elastically confined. As noted earlier, this value is far in excess of that observed in the laboratory ion-exchanged glass of the same composition (<350 MPa). Altering the boundary conditions of the simulations to better model the anisotropic boundary conditions of laboratory ion-exchanged glasses may change the system responses, both with regard to expansion and possibly structural features; therefore the results in this work should, perhaps, be interpreted cautiously. The next phase of investigation will include exploration of structural changes under anisotropic boundary conditions.

The primary intention of the present study was to examine the structural response of silicate networks immediately after accommodating

stuffing alkali ions. Ion-exchange temperature of 300 K was selected to largely exclude thermally-activated relaxations, thereby simplifying the interpretation of the resultant structures by excluding such relaxations. Although ion exchange at 300 K is inaccessible in the laboratory for these glass compositions, molecular dynamics simulations allow the use of such temperature and this control enables the segregation of immediate structural response and thermally-activated structural relaxations. Thus, these ion-exchange simulations at 300 K serve as a baseline for understanding the response of the glass structure undergoing the alkali accommodation procedure. Future investigation will specifically examine relaxation and structural evolution of ion-exchange stuffed structures at temperatures greater than 300 K. To briefly demonstrate the influence of performing ion exchange at elevated temperature, one simulation where exchange was performed at 623 K has been included here. The closeness of the results between ion-exchange stuffing simulations performed at 300 K and at 623 K, then compared at 300 K, showed few notable differences for the properties examined. The minor change of of RSD (Fig. 2) and small increase of molar volume (Table 2) indicate the possibility for some amount of thermally-activated relaxation. The subtlety of the property differences is contrary to what would be anticipated from a 300 K temperature difference in a laboratory ion-exchange experiment. However, while a laboratory ion-exchange experiment is typically carried out on time scales of minutes or hours, these simulations explore time scales of pico- or nanoseconds. It is possible that the brief observation time of the simulation heavily outweighs the effect of increasing the ion-exchange temperature by 300 K. Thus, whether performed at 300 K or 623 K, the structural features of ion-exchange accommodation appear similar over the simulation time scales used here.

It was not clear if irreversible mechanisms, or activated processes, of alleviating network strain, other than thermally-activated relaxations which have largely been omitted here by control of temperature, could or would be present in these simulations. Results of back-exchanged ($x_{\text{SwapBack100_npt_0}}$) simulations, for the temperature (300 K), time scales, and properties investigated here, suggest the deformation of the silicate network caused during accommodation of stuffing alkali ions is nearly completely elastic. That is, elastic behavior is observed upon removal of stuffing potassium ions for sodium ions, where all examined properties return to that of the as-formed host glass. The slight differences of the Si–O–Si bond angle and molar volume after back-exchange are within the error of these properties and would, perhaps, be unmentioned if not for the consistency of this trend between the three back-exchange simulations. This suggests the possibility exists for a subtle permanent deformation of the silicate network caused by accommodation of stuffing alkali, but based upon these observations its contribution to the accommodation process is minor at the temperature and time scales used here, if present.

A recent study by Tandia et al. [20] examined ion exchange in mixed sodium-potassium silicate glasses with 20 mol% total alkali content. Utilizing molecular dynamics simulations, stuffing potassium ions replaced the sodium ions in the various mixed alkali compositions and the resulting structures were examined. The as-formed configurations of the $x=23$ mol% sodium and potassium silicate compositions in the present study compare acceptably well against the 20 mol% Na_2O silicate and 20 mol% K_2O silicate glasses of Tandia et al. Somewhat lower molar volumes (~ 0.7 – 1.1 cm^3/mol lower) are observed by Tandia et al., accompanied by slightly higher alkali-oxygen mean coordination numbers (~ 1 higher for (Na)O and ~ 1 higher for (K)O). These minor differences are likely attributable to differences in simulation technique. The present glasses were formed by traditional molecular dynamics techniques and Tandia et al. used a hybrid Metropolis Monte Carlo-molecular dynamics technique. With 20 mol% stuffing K_2O in a sodium tetrasilicate host, Tandia et al. observe mean (K)O CN of ~ 9.2 , compared to mean (K)O CN of ~ 8.4 for the as-melted potassium tetrasilicate composition (estimated from Tandia et al. Fig. 4a and b, respectively), unchanged Q^n distribution, and a resultant LNDC near $1.3 \times 10^{-3}/(\text{mol}\% \text{K}_2\text{O})$. This single simulation by Tandia et al. of network accommodation

of stuffing alkali is in agreement with the accommodation process observed in the present study.

Other simulations by Tandia et al. are of as-melted sodium-potassium silicate compositions, which constitute different host structures than the sodium silicates hosts used in the present study, and are thus not equivalent to the stuffing simulations here. Comparison of the manner in which binary sodium silicate and mixed sodium-potassium silicate simulated glass networks accommodate stuffing potassium ions shows similarities and differences. When mixed sodium-potassium silicate host glasses were potassium-stuffed, Tandia et al. observed no net Q^n distribution changes and also lower molar volumes compared to the as-melted potassium tetrasilicate. This is similar to the observations made in this study of potassium-stuffed binary sodium silicate host glasses. For as-melted sodium-potassium silicate host glasses, where the sodium portion of the composition was replaced by stuffing potassium, Tandia et al. observe potassium-oxygen CN that is less than that of both the as-melted host glass and the as-melted potassium tetrasilicate. In contrast, the potassium-stuffed sodium silicate hosts in the present study showed potassium-oxygen CN near, or slightly greater than, the as-formed potassium silicate glass. Therefore, differences in the structural features of mixed sodium-potassium silicate hosts and the binary sodium silicate hosts partly modify the manner in which the stuffing potassium ions are accommodated by the silicate network.

For the sodium-potassium silicates, Tandia et al. explained the molar volume difference between ion-exchange stuffed glasses and their end-member potassium tetrasilicate glass by under-coordination of oxygen around potassium. While this may well be partly responsible for the resulting molar volumes of their ion-exchange stuffed glasses, based upon the observations in the present work, a second factor governing the resulting molar volume is the silicate network topology. Tandia et al. did not observe net Q^n distribution change, which is consistent with the findings here, but as noted earlier the difference of Q^n distributions between the as-formed end-member glasses is small. Thus, without additional information about the longer-range structures, such as ring size distributions, in the simulated glasses of Tandia et al., one can only speculate, but it is suspected that topological confinement limits the achievable molar volumes not only of the binary sodium silicates but also the mixed sodium-potassium silicates after ion-exchange stuffing.

Structural configurations of the simulated ion-exchange strengthened glasses in the present study are notably different than their as-formed potassium silicate counterparts, especially with regard to topological configuration, and are also notably different than their as-formed host with regard to oxygen coordination number around alkali and Si–O–Si bond angle. As such, one may conclude that the order parameters [23] of these configurations are unique and are likely inaccessible to fusion forming techniques. In agreement with the conclusion of Tandia et al. [20], ion-exchange stuffed structural configurations are examples of ‘forbidden glasses’ [24] for which no as-melted configuration can replicate in the absence of external forces on the system.

5. Conclusions

Molecular dynamics simulations of ion-exchange stuffed glasses were shown to have lower molar volume than their compositionally-equivalent, as-formed end-member glasses. Upon exchange of sodium ions with potassium ions in a sodium silicate host, stuffing potassium ions drove limited alkali site expansion obtaining mean (K)O coordination number similar to that found in the as-formed potassium silicate end-member glass. The topological configuration of the network remained that of the sodium silicate host glass and the resulting network strain was borne, in part, by intertetrahedral re-orientation. Removal of the stuffing potassium ions by replacement with sodium

ions appeared to restore the structural features of the network very close to those of the as-formed sodium silicate host glass.

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